

Solution.

Finding the missing number

407

Challenge. (Note: Two of the boxes contain a negative number).

43. $\boxed{17} \xrightarrow{+3} \boxed{20} \xrightarrow{\div 4} \boxed{5} \xrightarrow{\times 3} \boxed{15} \xrightarrow{+2} \boxed{17} \xrightarrow{-5} \boxed{12}$
 $\downarrow -4$
 $\boxed{24} \xrightarrow{+11} \boxed{13} \xrightarrow{+8} \boxed{5} \xrightarrow{\div 6} \boxed{30} \xrightarrow{\times 3} \boxed{10} \xrightarrow{+2} \boxed{8}$
 $\downarrow \div 2$
 $\boxed{12} \xrightarrow{\div 2} \boxed{6} \xrightarrow{\div 2} \boxed{3} \xrightarrow{+6} \boxed{9} \xrightarrow{-11} \boxed{-2} \xrightarrow{+8} \boxed{6}$
 $\downarrow \times 3$
 $\boxed{3} \xrightarrow{-7} \boxed{10} \xrightarrow{-8} \boxed{18} \xrightarrow{-8} \boxed{26} \xrightarrow{\times 2} \boxed{13} \xrightarrow{-5} \boxed{18}$
 $\downarrow \times 7$
 $\boxed{21} \xrightarrow{+9} \boxed{30} \xrightarrow{\times 2} \boxed{60} \xrightarrow{\div 5} \boxed{12} \xrightarrow{-8} \boxed{4} \xrightarrow{-3} \boxed{1}$
 $\downarrow \times 1$
 $\boxed{7} \xrightarrow{+7} \boxed{0} \xrightarrow{\times 8} \boxed{0} \xrightarrow{-4} \boxed{4} \xrightarrow{-4} \boxed{8} \xrightarrow{+7} \boxed{1}$
 $\downarrow \times 5$
 $\boxed{35} \xrightarrow{-8} \boxed{27} \xrightarrow{\div 3} \boxed{9} \xrightarrow{\times 2} \boxed{18} \xrightarrow{+2} \boxed{20} \xrightarrow{\times 5} \boxed{100}$
 $\downarrow \div 4$
 $\boxed{9} \xrightarrow{\div 2} \boxed{18} \xrightarrow{\times 3} \boxed{6} \xrightarrow{\times 3} \boxed{2} \xrightarrow{\div 8} \boxed{16} \xrightarrow{-9} \boxed{25}$
 $\downarrow + 1$
 $\boxed{10} \xrightarrow{+2} \boxed{12} \xrightarrow{+3} \boxed{15} \xrightarrow{+3} \boxed{18} \xrightarrow{\div 2} \boxed{9} \xrightarrow{\times 7} \boxed{63}$
 $\downarrow + 9$
 $\boxed{5} \xrightarrow{\div 2} \boxed{10} \xrightarrow{\times 5} \boxed{2} \xrightarrow{-6} \boxed{8} \xrightarrow{\div 3} \boxed{24} \xrightarrow{\div 3} \boxed{72}$
 $\downarrow + 1$
 $\boxed{6} \xrightarrow{-8} \boxed{-2} \xrightarrow{+10} \boxed{8} \xrightarrow{\times 2} \boxed{16} \xrightarrow{+4} \boxed{20} \xrightarrow{-6} \boxed{14}$
 $\downarrow + 1$
 $\boxed{6} \xrightarrow{\times 3} \boxed{2} \xrightarrow{\div 4} \boxed{8} \xrightarrow{+3} \boxed{5} \xrightarrow{\div 9} \boxed{45} \xrightarrow{\times 3} \boxed{15}$
 $\downarrow \times 4$
 $\boxed{24} \xrightarrow{-7} \boxed{17} \xrightarrow{-7} \boxed{10} \xrightarrow{\div 5} \boxed{2} \xrightarrow{+3} \boxed{5} \xrightarrow{+3} \boxed{8}$
 $\downarrow \times 1$
 $\boxed{7} \xrightarrow{\div 2} \boxed{14} \xrightarrow{+4} \boxed{10} \xrightarrow{\div 3} \boxed{7} \xrightarrow{-9} \boxed{16} \xrightarrow{\times 2} \boxed{8}$
 $\downarrow \times 3$
 $\boxed{21} \xrightarrow{-1} \boxed{20} \xrightarrow{+6} \boxed{26} \xrightarrow{\div 2} \boxed{13} \xrightarrow{+1} \boxed{14} \xrightarrow{\div 2} \boxed{7}$

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N 3.3: Adds & subtracts whole numbers ... & multiplies & divides by one digit whole numbers ..
 N 4.2: Understands the ... connections between the four operations on whole ... numbers, and
 uses this understanding to choose appropriate operations ...